

CHAPTER 6 - The normal distribution and other continuous distributions

SDS188 - Group 3

2026

1 Normal distributions

Type of variable: Continuous

$$X \sim N(\mu, \sigma^2)$$
$$-\infty < x < \infty$$

1.1 Properties of the normal distribution

1.1.1 Parameters

- μ : the average of the X values
- σ^2 : the variance of the X values

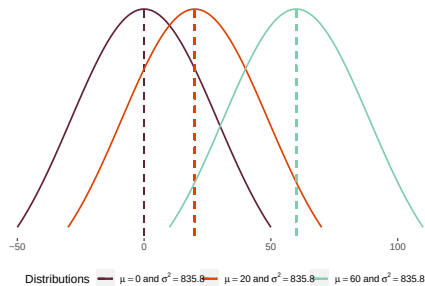
1.1.2 Density function

$$f(X) = \frac{e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2}}{\sqrt{2\pi}\sigma}$$

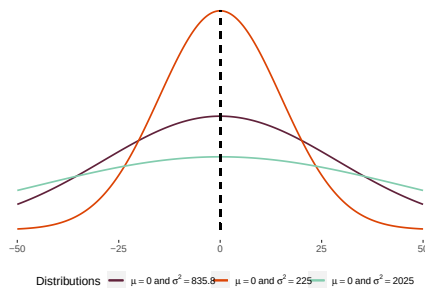
* π refers to the mathematical constant 3.14159...

1.1.3 The effect of the parameters on the shape of the distribution

Change in μ with σ constant- shifts the curve and keeps the shape



Change in σ with μ constant - changes the shape of the curve and stays on the same place



1.2 The standard distribution

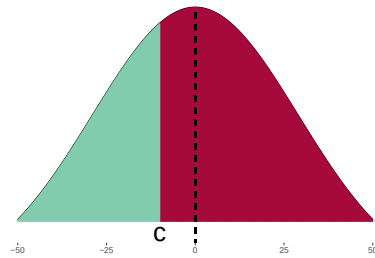
If $X \sim N(\mu, \sigma^2)$

$\longrightarrow Z \sim N(\mu = 0, \sigma^2 = 1)$

$$f(Z) = \frac{e^{-\frac{1}{2}(Z)^2}}{\sqrt{2\pi}}$$

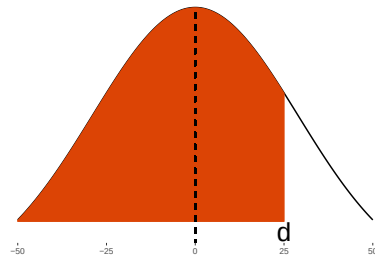
1.3 Calculate probabilities for the normal distribution

$$P(X > c)$$



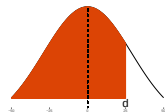
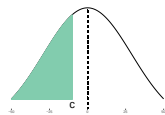
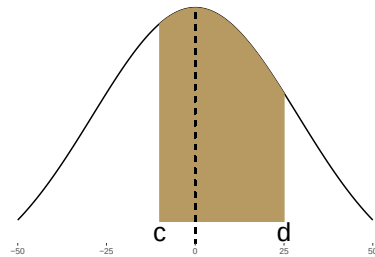
$$P(X > c) = 1 - P(X \leq c)$$

$$P(X < d)$$



$$P(X < d)$$

$$P(c < X < d)$$



$$P(c < X < d) = P(X < d) - P(X < c)$$

1.3.1 Steps

1. Draw problem in curve in terms of X

2. Transform X to $Z \rightarrow Z = \frac{X-\mu}{\sigma}$

$$P(X \leq d) \rightarrow P(Z \leq \frac{d-\mu}{\sigma})$$

3. Methods:

- **Z-table:** Find z in rows and columns of table with cumulative standard normal probabilities en read probability
- **HP calculator:** z  
- **Excel:** =NORM.S.DIST(z ,TRUE) of =NORM.DIST(x,μ,σ ,TRUE)

The table with cumulative Z probabilities is on the next two pages:

Z	Cumulative Probabilities									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-6.0	0.00000001									
-5.5	0.00000019									
-5.0	0.000000287									
-4.5	0.000003398									
-4.0	0.000031671									
-3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
-3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
-3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
-3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
-3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
-3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
-3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
-3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
-3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
-3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Z	Cumulative Probabilities									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99897	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4.0	0.999968329									
4.5	0.999996602									
5.0	0.999999713									
5.5	0.999999981									
6.0	0.999999999									

1.3.2 Calculate x given a probability

Let x_0 be a unknown value of X .

$P(X < x_0)$:

$$P(X < x_0) = P(Z < z_0)$$

Determine the value of z_0 by using the following methods:

- Z -table:

Find the closest probability in the table and read off the corresponding value of Z .

- HP-calculator:

$$P(Z < z_0) \quad \text{[HP icon]} \quad \text{[INV]} \quad \text{[Z=P]} \quad \text{[3]} \quad \text{[n!]}$$

- Excel:

=NORM.S.INV($P(Z < z_0)$) or =NORM.INV($P(X < x_0)$, μ , σ)

$P(X > x_0)$:

$$P(X > x_0) = P(Z > z_0) = 1 - P(Z < z_0)$$

Determine the value of z_0 by using the following methods:

- Z -table:

Find the closest probability in the table and read off the corresponding value of Z .

- HP-calculator:

$$P(Z < z_0) \quad \text{[HP icon]} \quad \text{[INV]} \quad \text{[Z=P]} \quad \text{[3]} \quad \text{[n!]}$$

- Excel:

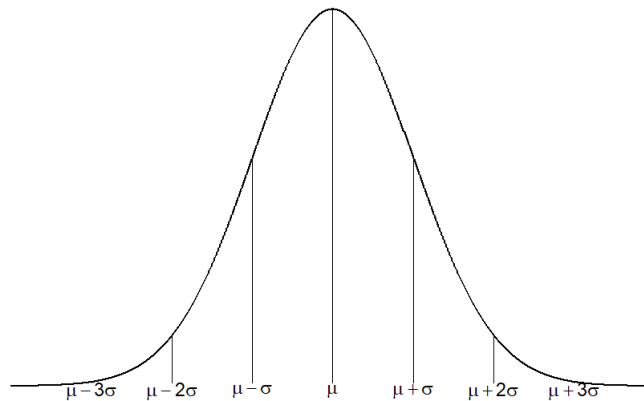
=NORM.S.INV($P(Z < z_0)$) or =NORM.INV($P(X < x_0)$, μ , σ)

Transform Z back to X where $x_0 = \mu + (z_0 \times \sigma)$

1.4 Evaluation of data for normality

We can evaluate if any observed continuous variable is theoretically normally distributed

$$X \sim N(\mu, \sigma^2)$$



Compare observed data with theoretical properties of the normal distribution

I. Step 1. Theoretical property: Symmetry by means of measures

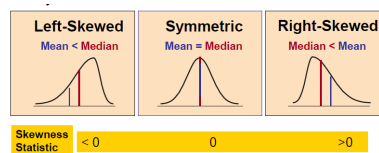
A. Calculate mean, median, mode (if possible) and variance of data.

According to the normal distribution, the mean, median and mode are equal

If $\bar{x} = Q_2 = \text{modus}$ or $\bar{x} \approx Q_2 \approx \text{modus}$, the data is symmetrically distributed and possibly normally distributed or approximately normally distributed

If $\bar{x} \neq Q_2 \neq \text{modus}$, the data is not symmetrically distributed and is either skew to the right or skew to the left.

- $\bar{x} < Q_2$ - skew to the left
- $\bar{x} > Q_2$ - skew to the right

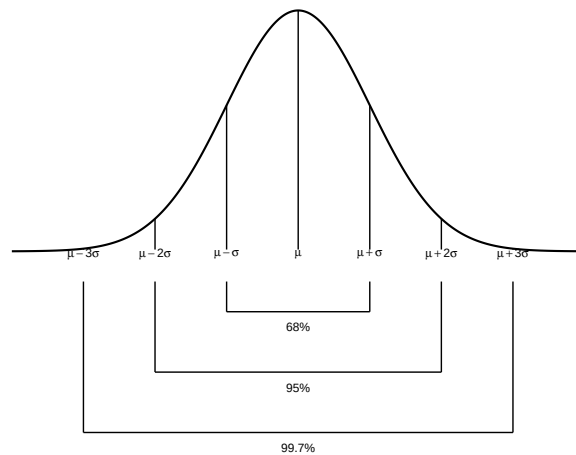


B. Use $(\bar{x})$ and standard deviation (s) to apply empirical rule

What proportion of data lies between $(\bar{x} - s)$ and $(\bar{x} + s)$? According to the empirical rule, the proportion must be 68%.

What proportion of data lies between $(\bar{x} - 2s)$ and $(\bar{x} + 2s)$? According to the empirical rule, the proportion must be 95%.

What proportion of data lies between $(\bar{x} - 3s)$ and $(\bar{x} + 3s)$? According to the empirical rule, the proportion must be 99.7%.



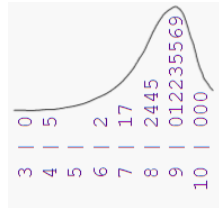
II. Step 2. Theoretical property: Symmetry and bell shape by means of graphs

A. For small datasets:

i. Stem and leaf diagram

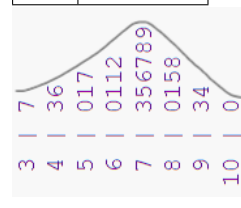
Skew to the left

3	0
4	5
5	
6	2
7	17
8	2445
9	012235569
10	000



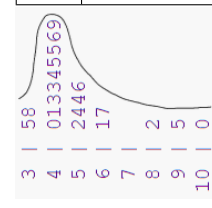
Symmetric

3	7
4	36
5	017
6	0112
7	356789
8	0158
9	34
10	0

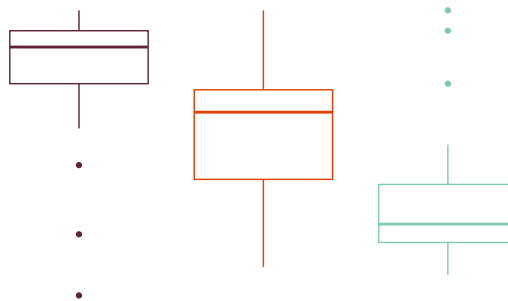


Skew to the right

3	58
4	013345569
5	2446
6	17
7	
8	2
9	5
10	0



ii. Boxplot (can also use 5-point summary)



 Skeef na links
  Simmetries
  Skeef na regs

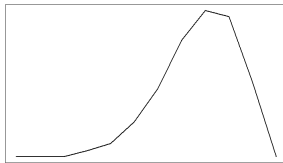
Left-Skewed	Symmetric	Right-Skewed
Median - $x_{\min}$ >	Median - $x_{\min}$ $\approx$	Median - $x_{\min}$ <
$x_{\max}$ - Median	$x_{\max}$ - Median	$x_{\max}$ - Median
$Q_1 - x_{\min}$ >	$Q_1 - x_{\min}$ $\approx$	$Q_1 - x_{\min}$ <
$x_{\max} - Q_3$	$x_{\max} - Q_3$	$x_{\max} - Q_3$
Median - Q_1 >	Median - Q_1 $\approx$	Median - Q_1 <
Q_3 - Median	Q_3 - Median	Q_3 - Median

Symmetric $\rightarrow X \sim N(\mu, \sigma^2)$

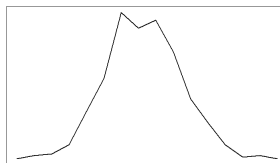
B. For large datasets:

i. **Relative frequency polygon:**

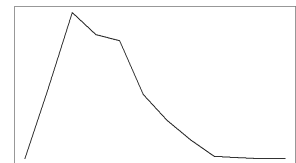
Skew to the left



Symmetric

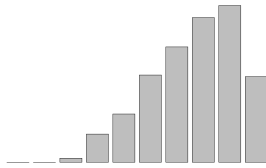


Skew to the right

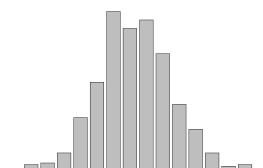


ii. **Histogram:**

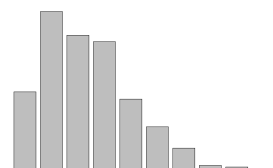
Skew to the left



Symmetric



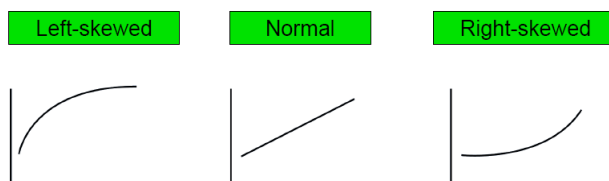
Skew to the right



III. Step 3. Theoretical property: linear relationship between Z and X values

A. **Normal probability plot**

1. Sort data in ascending order
2. Create index j
3. Calculate $\alpha_j, \alpha_j = \frac{j}{n+1}$
4. Calculate corresponding Z values for each α_j
 α_j are probabilities $P(Z < z_0)$, find z_0
5. Draw scatterplot with X values on y axis and Z values on x axis
6. Evaluate normality by looking at the shape of the plot/distribution of points



Determine conclusion following the steps

IV. In summary:

Calculate:

- Mean
- Median
- Mode
- 1st and 3rd quartiles
- Minimum and maximum
- Variance and standard deviation
- Proportions for empirical rule B.
- Frequency distribution B.
- Probabilities α_j A.

Test for:

- Symmetry
 - Mean = median A.
 - Five-point summary II.(A.)ii
 - Empirical rule B.
 - Graphs II.(A.)i, II.(A.)ii, II.(B.)i, II.(B.)ii
- Bell shape
 - Histogram II.(B.)ii
 - Relative frequency polygon II.(B.)i
 - Stem and leaf diagram II.(A.)i
- Linear relationship between X and Z
 - Normal probability plot A.

1.5 Normal approximation to binomial distribution

$$X \sim \text{binomial}(n, \pi)$$

$$P(X = x|n, \pi) = \binom{n}{x} \pi^x (1 - \pi)^{n-x}$$

As n increases, the calculations for $P(X = x|n, \pi)$ become more difficult.

For example:

$$x = 5, \pi = 0.2$$

For small n ($n = 12$):

$$\begin{aligned} P(X = 5|n = 12, \pi = 0.2) &= \binom{12}{5} (0.2)^5 (0.8)^7 \\ &= 0.0531 \end{aligned}$$

For large n ($n = 500$):

$$\begin{aligned} P(X = 5|n = 500, \pi = 0.2) &= \binom{500}{5} (0.2)^5 (0.8)^{495} \\ &= 255244687600 \times 0.00032 \times 0.000000000... \\ &= 0.000000000... \\ &\approx 0 \end{aligned}$$

Solution! Normal probabilities can be used to calculate binomial probabilities. This approximation can only be used if:

- $n\pi > 5$
- $n(1 - \pi) > 5$

If $n\pi > 5$ and $n(1 - \pi) > 5$:

$$X \sim \text{binomial}(n, \pi) \longrightarrow X_a \sim N(\mu, \sigma^2)$$

where X_a is the approximation for X

$$\mu = n\pi$$
$$\sigma^2 = n\pi(1 - \pi)$$

BUT

Binomial distribution	Normal distribution
• Only integer outcomes. • $P(X = x) \geq 0$	• Outcome within $(-\infty, +\infty)$. • $P(X = x) = 0$

Continuity correction:

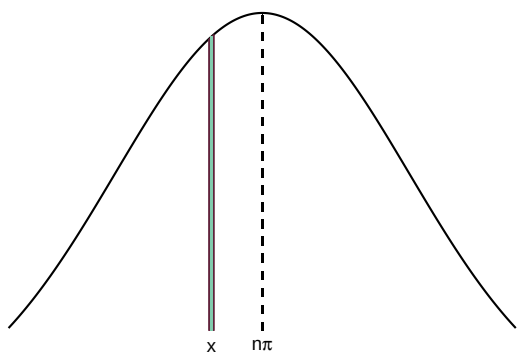
$$P(X = x) \approx P(x - 0.5 < X_a < x + 0.5)$$

$$P(X \leq x) \approx P(X_a < x + 0.5)$$

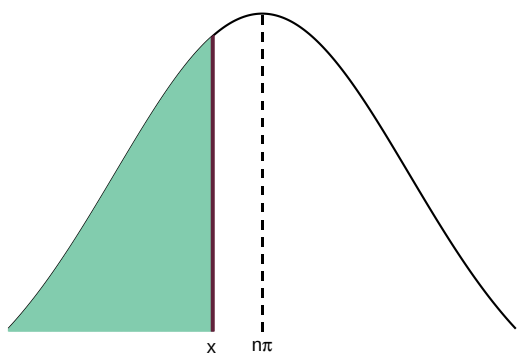
$$P(X \geq x) \approx P(X_a > x - 0.5)$$

$$P(X = x) \implies P(\dots X_a \dots) \implies P(\dots Z \dots)$$

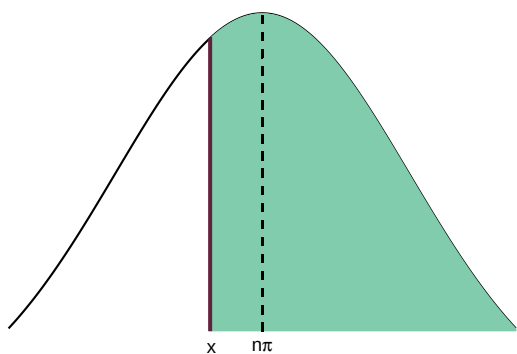
$$\underline{P(X = x) \approx P(x - 0.5 < X_a < x + 0.5)}$$



$$\underline{P(X \leq x) \approx P(X_a < x + 0.5)}$$



$$\underline{P(X \geq x) \approx P(X_a > x - 0.5)}$$



2 Uniform distribution

Type of variable: Continuous

$$X \sim \text{uniform}(a, b)$$

$$a \leq x \leq b$$

2.1 Properties of uniform distribution

2.1.1 Parameters

a is the minimum value of X

b is the maximum value of X

2.1.2 Density function

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq X \leq b \\ 0 & \text{otherwise} \end{cases}$$

2.1.3 Measures

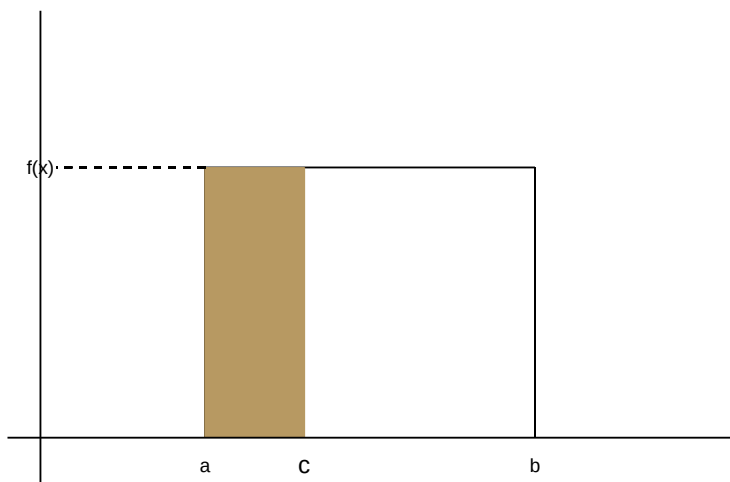
$$\mu = \frac{a+b}{2}$$

$$\sigma = \sqrt{\frac{(b-a)^2}{12}}$$

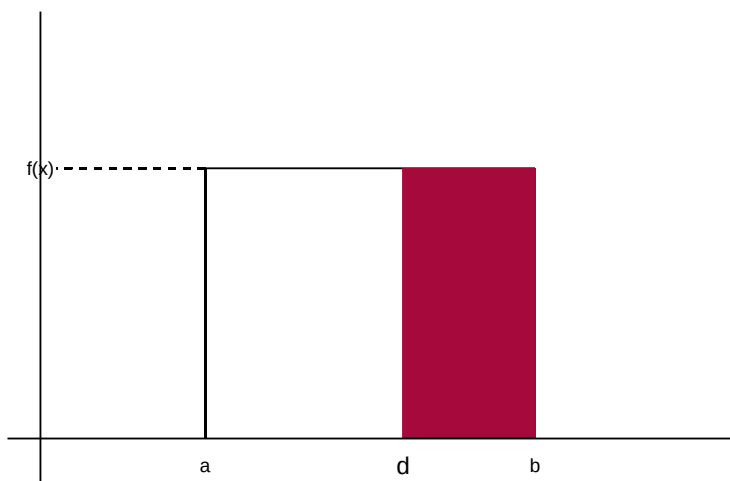
2.1.4 Probabilities

Method A:

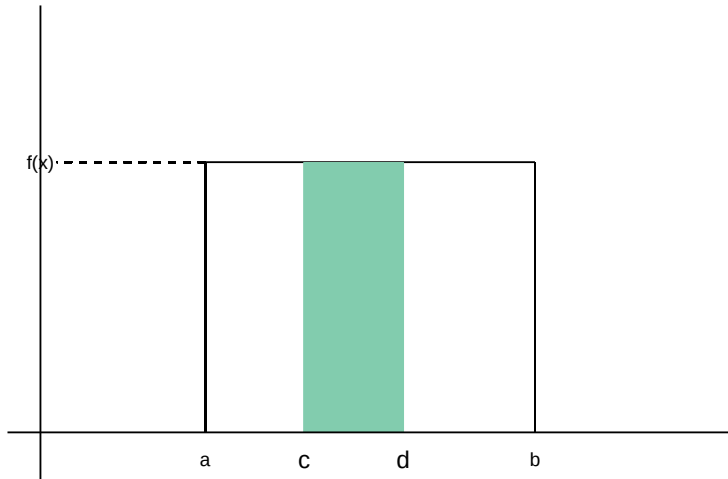
$$P(X < c) = (c - a) \times f(x)$$



$$P(X > d) = (b - d) \times f(x)$$

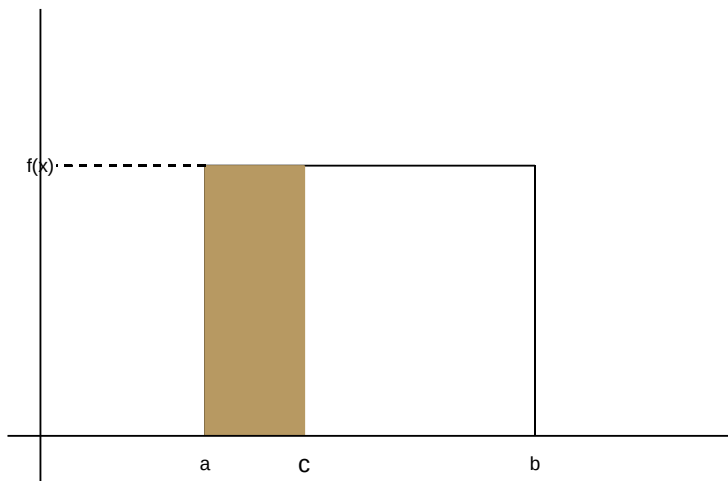


$$P(c < X < d) = (d - c) \times f(x)$$

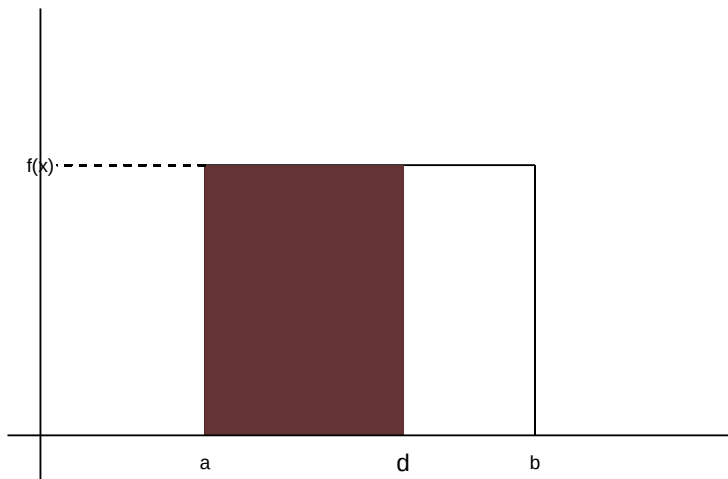


Method B:

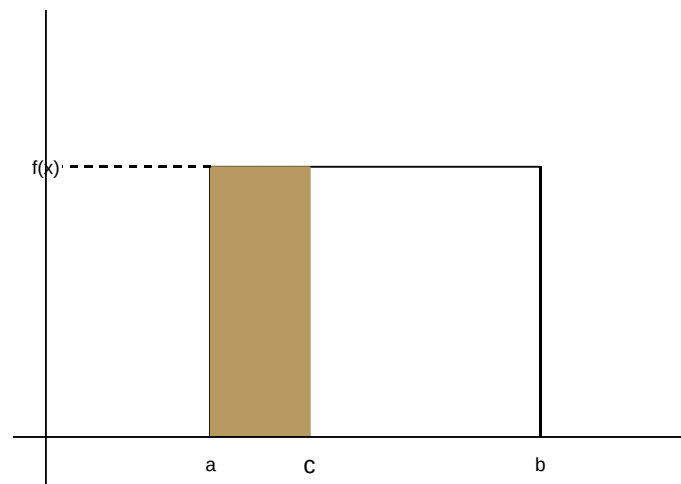
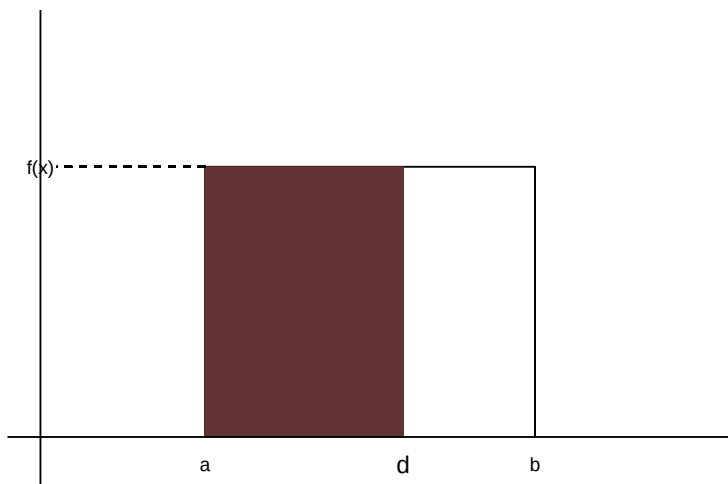
$$P(X < c) = \frac{c-a}{b-a}$$



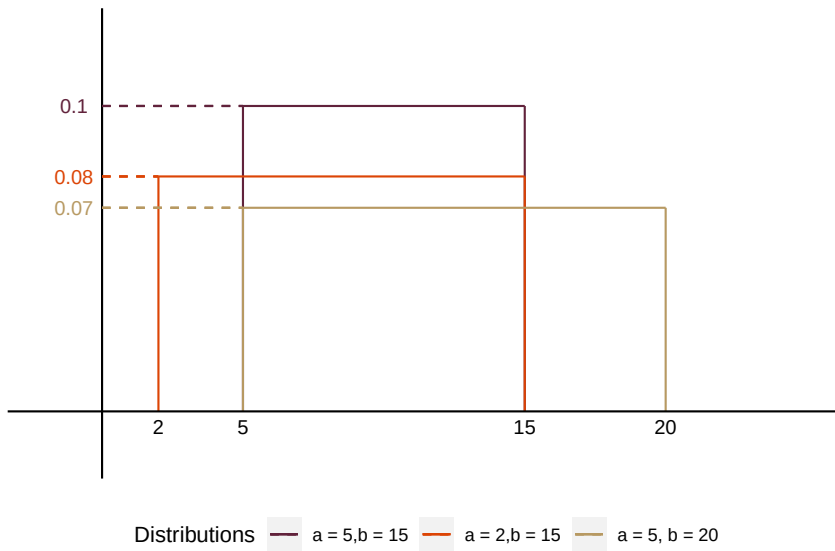
$$P(X > d) = 1 - P(X < d) = 1 - \frac{d-a}{b-a}$$



$$P(c < X < d) = P(X < d) - P(X < c) = \frac{d-a}{b-a} - \frac{c-a}{b-a}$$



2.1.5 The effect of the parameters on the shape of the distribution



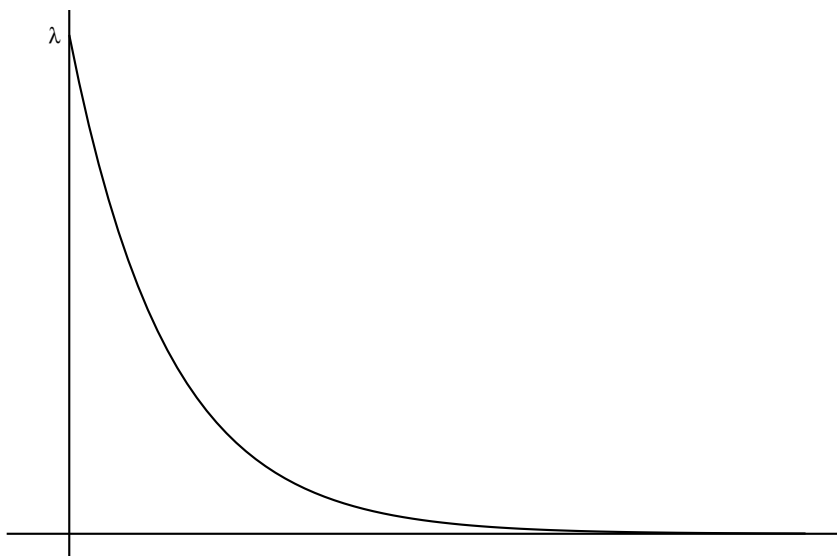
3 Exponential distribution

Type of variable: Continuous

$$X \sim \exp(\lambda)$$

$$0 \leq x < \infty$$

$$x \geq 0$$



3.1 Properties of the exponential distribution

3.1.1 Parameters

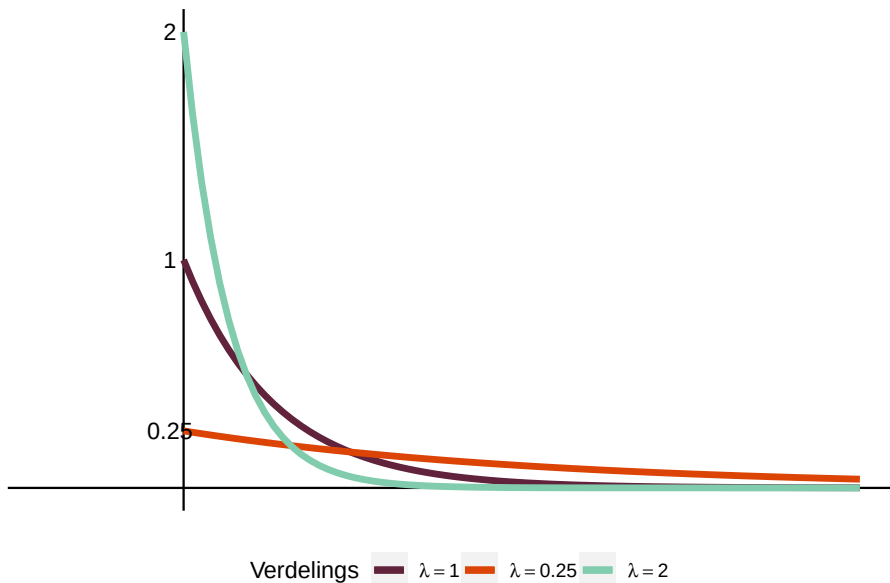
λ refers to the average rate per unit of time/area of opportunity.
This can be the same λ as that of the Poisson distribution.

3.1.2 Density function

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{as } X \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

3.1.3 The effect of the parameters on the shape of the distribution

Regardless of the value of λ , the curve will always be skew to the right. The decrease in λ will shift the curve the right.



3.1.4 Measures

$$E(X) = \frac{1}{\lambda}$$
$$Var(X) = \frac{1}{\lambda^2}$$

3.1.5 Probabilities

$$P(X \leq x) = 1 - e^{-\lambda x} \quad \text{for} \quad x > 0$$
$$P(X \geq x) = 1 - (1 - e^{-\lambda x}) = 1 - 1 + e^{-\lambda x} = e^{-\lambda x}$$

4 Questions

4.1 Question 1

A study of the time spent shopping in a supermarket for 20 specific items showed an approximately uniform distribution between 10 minutes and 20 minutes.

1. What is the probability that the shopping time will be between 12 and 18 minutes?
2. What is the probability that the shopping time will be less than 17 minutes?
3. What is the mean and standard deviation of the shopping time?

4.2 Question 2

Given an exponential distribution with $\lambda = 3$. Calculate the following:

1. $P(X > 0.45)$
2. $P(0.2 < X < 0.5)$
3. x_0 if $P(X > x_0) = 0.65$

4.3 Question 3

A trucking company determined that the distance travelled per truck per year is normally distributed, with a mean of 30 thousand miles and a standard deviation of 10 thousand miles.

1. What percentage of trucks can be expected to travel between 18 thousand and 30 thousand miles in a year?
2. What percentage of trucks can be expected to travel either less than 20 thousand or more than 50 thousand miles in year?
3. How many miles will be travelled by at least 60% of the trucks?

4.4 Question 4

The mean number of automobiles entering a mountain tunnel per hour is 30. An excessive number of cars entering the tunnel during a brief period of time produces a hazardous situation. If a car has just entered the mountain tunnel, what is the probability that the next car enters the mountain tunnel after 20 minutes?

4.5 Question 5

One year consumers spent an average of \$24 on a meal at a restaurant. Assume that the amount spent on a restaurant meal is normally distributed and the standard deviation is \$6.

1. What is the probability that a randomly selected person spent more than \$25?
2. What is the probability that a randomly selected person spent between \$8 and \$20?
3. Between what two values, symmetrically distributed around the mean, will be the middle 72% of the amounts of cash spent fall?

4.6 Question 6

Suppose that Y has a binomial distribution with $n = 100$ and $\pi = 0.1$. Use the normal distribution to calculate $P(Y > 65)$.

4.7 Question 7

According to a recent study, 88% of retail brands use two or more social media channels for business. Consider a sample of 10 businesses is selected. Use the normal distribution to approximate the following probabilities.

1. What is the probability that 7 businesses use two or more social media channels?
2. What is the average and standard deviation of the number of businesses that use two or more social media channels?
3. What is the probability that at least 8 businesses use two or more social media channels?

4.8 Question 8

Suppose you sample one value from a uniform distribution with $a = 0$ and $b = 20$.

1. What is the probability that the value will be between 12 and 19?
2. What is the probability that the value will be between 2 and 8?
3. What is the mean?
4. What is the standard deviation?

4.9 Question 9

Customers arrive at a checkout counter in a department store according to a Poisson distribution at an average of seven per hour. During a given hour, what is the probability that a client

1. Arrives after 5 minutes?
2. Arrives between 12 and 20 minutes?
3. Arrives within half an hour?
4. If the probability that a client arrives within a specific time is 87%, what is the specific time?

4.10 Question 10

Given a normal distribution and given information, determine the following:

1. If $\mu = 25$, $\sigma = 2.9$, find $P(X < 31)$?
2. If $\mu = 9$, $\sigma = 7.05$, find $P(X < 3)$?
3. If $\mu = 13$, $\sigma = 0.9$, $P(X > x) = 0.4$, find x ?